

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

Contents

DELIVERABLES and DEADLINES	2
PROJECT DETAILS.....	2
Base Technical Requirements and Rules.....	2
RESTful Service Definitions.....	3
Main Objects And Properties	7
GRADING	8
Grading Rules For Delivery-1	8
Grading Rules For Delivery-2	9
Grading Rules For Delivery-3	9
Penalties	9

- This assessment must be prepared and finalized individually or as a team.
- You must **upload** your project into LMS as an archive file.

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

DELIVERABLES and DEADLINES

You'll work on your project step by step. The required features for each stage are listed below. After completing the relevant parts, upload your project to the LMS by the specified deadline. Please do not include any code other than what is required, and make sure the project you upload is working properly.

DELIVERABLES	Weight	Deadline	Expectation
initial project setup & domain model	%5	November 9, 2025	• SpringBoot project with related dependencies must be created • Domain classes and relations must be created(not entities) • A UML class diagram to show object relations is provided
database&JPA config&testing	%15	November 30, 2025	• Database configuration must be completed • Entity definitions must be completed. • The application must insert initial data to the database when it is started (you can use insert scripts or SpringBoot Runners) • All Spring Data JPA repositories must be created • All service layer codes needed by the controller layer (Restful services) must be completed. • Junit tests for all service methods must be developed
Restful services&testing &demo	%20	December 21, 2025	All requirements defined in this document must be completed.

PROJECT DETAILS

Suppose that you are working in a company as a software developer. Your company is developing a **Car Rental Application**. A web application and a mobile application will be developed. Your team is responsible for developing all required Restful services which are required for web/mobile applications.

Base Technical Requirements and Rules

1. Create a Spring Boot web application. Use JPA and Spring Data JPA. You are not allowed to use pure JDBC and Spring JdbcTemplate. But you can use native SQL in the Data JPA Repositories.
2. Generate an API documentation for your Restful services using OpenAPI (Swagger). Be careful, you will not develop a web application, you will use Swagger and it will produce the related API documentation.
3. You are not allowed to use Entity Objects for the input and output of Restful services(in Controller layer). Your services should only accept and return Data Transfer Objects(DTO). You can use projection methods or a mapping library like mapstruct (<https://mapstruct.org>) for such transformations.
4. You are free to choose any database, but please **avoid using any database-specific commands in your project**.
5. Switching to another database must be done easily by changing only application.properties file and the dependencies.

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

6. You must initialize your database automatically with at least one record in each table. You are free to select any method you like for initialization.
7. Draw a UML class diagram for your model classes.
8. Develop separate service classes for each entity. For example: ReservationService, CarService, MemberService etc.
 - a. All service classes should contain all base methods for related Entity. For example your MemberService should contain methods for creating, reading a member and returning all members
 - b. If you need, you can add extra methods for updating entity.
 - c. You should also add any other required methods according to the business requirements(see below)
9. Develop separate controller classes according to the RESTful Services requirements (listed below). For example: ReservationController, CarController, CarServiceController, MemberController etc
10. Use a 3-layer architecture: Controller-Service-Repository. You can call Service beans from the Controller, and you can call Repository beans from the Service beans. However, you cannot call Repository classes directly from RestController classes
11. Add Junit tests for testing all of your service methods
12. Apply resource naming rules.
13. It is forbidden to use FetchType.EAGER
14. It is forbidden to use CascadeType.ALL and CascadeType.REMOVE. You can use the other cascade types depending on your requirements.

RESTful Service Definitions

Develop all necessary Restful services to satisfy all the UI service requirements listed below.

#	Service Name	Service input/output parameters and business rules
1	Search available cars	Input parameters: • category (optional) • transmission type (optional) • daily price range (optional) • pick-up date (required) • drop-off date (required) • number of seats (optional) • pickup location (required) Return Value : available car list. Business Rules: • Search for cars that match the entered criteria. Don't forget to check availability of cars: ○ Car statu must be "Available" (Check car's statu) ○ Car must not be booked between the specified dates(Check all active reservations). • Be careful when creating your query: if a parameter is left blank, do not add it to your query

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

		Return Http Status Codes: • 200 => if available cars are found • 404 => if there is no available cars
2	Make a reservation	Input parameters: • Car barcode number • Pick-up date and time • Drop-off date and time • Member id • Pick-up location code • Drop-off location code • Extra list Return Value : • Reservation number • Pick-up date-time • Drop-off date-time • Pick-up location code and name • Drop-off location code and name • Total amount • Member id • Member name Business Rules: Before making reservation you should check the car status again 1. If status is not "Available" you can't complete the reservation. You should send 406-Not Acceptable status code. In this case don't create & save any Reservation object. 2. If car status is Available you can complete reservation and send 200 Http status code. Following requirements must be provided: a. Generate a unique 8 digits reservation number in the Service layer. b. Set Reservation statu ACTIVE c. Assign reservation create date as sysdate d. Assign pick-up date and drop-off date e. Compute total amount : dayCount * car daily price + extra list prices f. Don't save total amount into database, it must be calculated when requested Return Http Status Codes: • 200 – if the reservation is completed successfully • 206 – if selected car is not available

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

3	Get all rented cars	Input parameters: none Return Value : List all cars that are currently rented (The reservation must be in Active status on the system date) . Each object should contain following values: • Brand • Model • Car type • Transmission type • Barcode number • Reservation Number • Member name • Drop-off date and time • Drop-off location • Reservation day count Business Rules: none Return Http Status Codes: • 200 => OK • 404 => if no rented cars
4	Add an extra to a reservation	Input parameters: • Reservation number • Code for extra service/equipment Return Value : boolean Business Rules: • Return true if adding process is completed successfully. • Return false if extra is already added or code is not found Return Http Status Codes: • 200 – OK – if extra is added successfully • 404 – Not Found – if selected extra is not found • 500 – if an exception thrown

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

5	Return the car	Input parameters: • Reservation number Return Value : boolean Business Rules: • Return true if vehicle return is completed successfully. ○ Update reservation as follows: ▪ Status=COMPLETED ▪ Return date = system date ○ If dropOffLocation of reservation is different from location of the car then update location of the car. • Otherwise, do nothing and return false Return Http Status Codes: • 200 – OK – if the car return is completed successfully • 404 – Not Found – if the car(or reservarion) is not found • 500 – if an exception is thrown
6	Cancel a reservation	Input parameters: • Reservation number Return Value : boolean Business Rules: • Return true if the reservation is cancelled successfully. Then update the reservation status as CANCELLED • Return false if the reservation is not cancelled successfully Return Http Status Codes: • 200 – OK – if the reservation is cancelled successfully • 404 – Not Found – if reservation number is not found • 500 – if an any exception thrown
7	Delete a car	Input parameters: • Car barcode number Return Value : boolean Business Rules: • If car has been used for any reservation, do not delete the car and return "false". • Otherwise, you can delete the car. Return Http Status Codes: • 200 – OK – if delete operation is completed successfully • 404 – Not Found – if car is not found • 406 – Not Acceptable – if car is used in a reservation • 500 – if an exception is thrown

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

8	Delete a reservation	Input parameters: • Reservation number Return Value : boolean Return Http Status Codes: • 200 – OK – when deletion is successful • 404 – Not Found – if no reservation is found • 500 – if an exception is thrown Rules: Do not use Cascade.Remove on Reservation. When a reservation is removed, do not remove the related car and member. You must first disassociate them from the reservation.
---	-----------------------------	--

Main Objects And Properties

The classes and properties listed below are only a rough draft. For the actual design, please carefully review the service requirements defined in this document. You may need to add new classes or properties

Member	All members can search the catalog, as well as reserve, pick-up, and return a car. Members can have • Name • Address • Email • Phone • Driving license number A member can reserve more than one cars
Reservation	Responsible for managing reservations for a car. Reservation may have • Reservation number (must be autogenerated 8 digit string) • Creation date • Pick-up date and time • Drop-off date and time • Pick-up location • Drop-off location • Return date • Status • Member • Car • Extras • ... Reservation can have different status: ACTIVE, COMPLETED, CANCELLED A car can be used for another reservation

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

Location	The car rental system has multiple locations, each location has attributes like code, 'Name' to distinguish it from any other locations. Each location has a code and a name. Such as: • 1-Istanbul Airport • 2-Istanbul Sabiha Gökçen Airport • 3-Istanbul Kadıköy • 4-Izmir City Center
Car	The basic building block of the system. Every car has a barcode, license plate number, number of seats, brand, model, mileage, transmission type, dailyPrice, category, location etc. Some of the categories are: • Compact car • Mid-size car • Large car • Station wagon • Luxury car • Minivans • SUVs Car status can be one of them: • AVAILABLE • BEING_SERVICED Transmission types: • Automatic • Manual
Extra Service/Equipment	Stores details about the various types of extras that members can add to their reservation. Some of them are: • Additional driver • Infant car seat • Car seat • GPS • Snow chains • Snow tires • Wifi router Each extra has a name and price. Price is not daily.

GRADING

Grading Rules For Delivery-1

	TASK	Points
1	SpringBoot project with related dependencies must be created	30
2	Domain classes and relations must be created	40
3	UML class diagram to show object relations is provided	30

CS393 – Project Assignment

Assignment Date : 14.10.2025

Deadlines

Delivery 1:	November 9, 2025	23.59 PM
Delivery 2:	November 30, 2025	23.59 PM
Delivery 3:	December 21, 2025	23.59 PM

Grading Rules For Delivery-2

	TASK	Points
1	Database configuration	10
2	Entity definitions	20
3	Initial data insertion	10
4	Spring Data JPA repositories	20
5	Service layer development	20
6	JUnit tests for all service methods	20

Grading Rules For Delivery-3

	Tasks	Points
1	API Documentation - swagger	12
2	Using DTO for input and output in the service and controller level	12
3	Search available cars	12
4	Make a reservation	12
5	Get all rented cars	10
6	Add an extra to a reservation	9
7	Return the car	8
8	Cancel a reservation	7
9	Delete a car	8
10	Delete a reservation	10

Penalties

1	Project is not a SpringBoot project	Failure Reason
2	There are database-specific codes	[-50,-1]
3	Using CascadeType.REMOVE or CascadeType.ALL	[-50,-1]
4	Using FetchType.Eager	[-50,-1]
5	not following resource naming rules	[-20,-1]
6	not using SpringDataJPA repositories	Failure Reason
7	No separate Controller classes for base entities	[-50,-1]
8	Layered architecture not used. (3 layers: Controller-Service-Repository)	[-50,-1]
9	Using pure JDBC or JdbcTemplate	Failure Reason